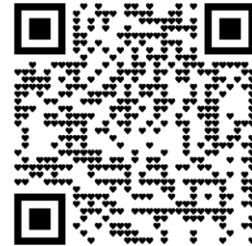
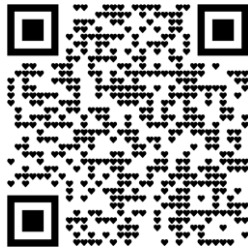


Setting up AWS Client VPN



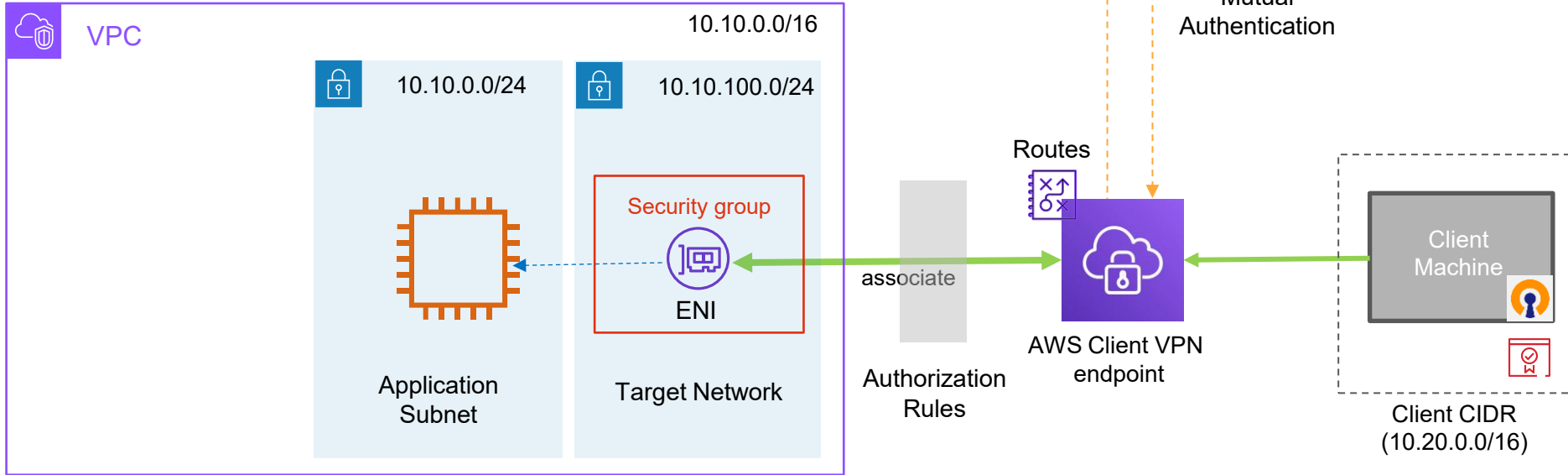
www.awswithchetan.com



<https://www.youtube.com/@AWSwithChetan>

Our Architecture

Make sure that VPC CIDR and Client CIDR doesn't overlap with your local LAN address

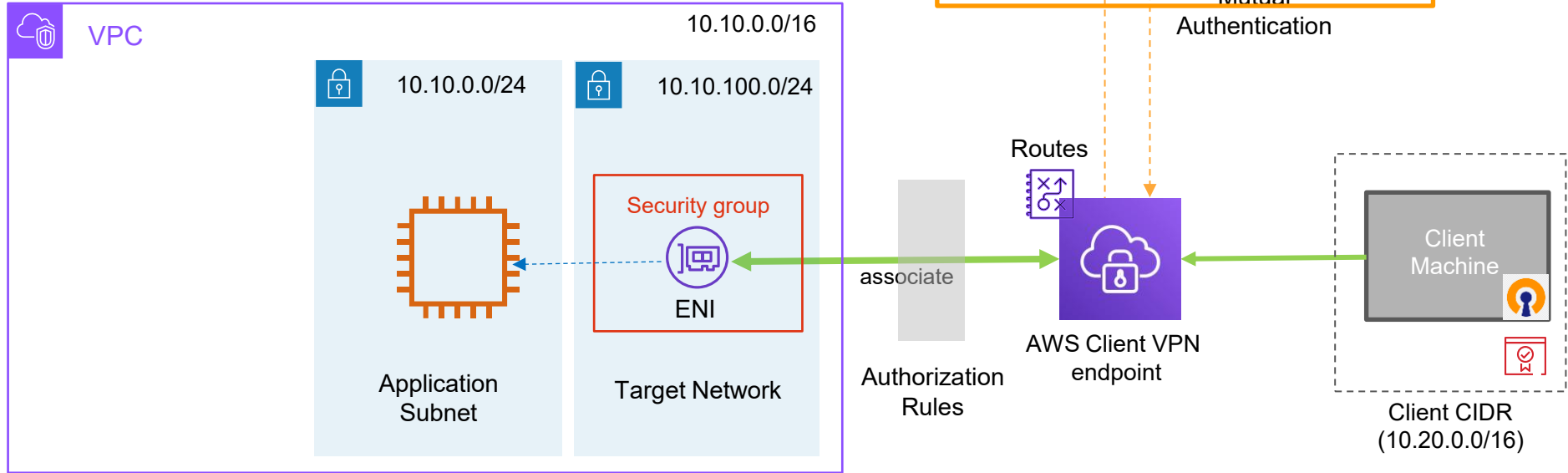


Exercise Steps

1. Create certificates and keys for Mutual Authentication and import into ACM
2. Setup VPC, Subnets and Security Groups
3. Create Client VPN Endpoint
4. Associate Target Network Subnet, Create Routes and Create Authorization rules in Client VPN
5. Download Client VPN Configuration file and Configure VPN Client (local machine)
6. Connect using VPN Client from your local machine
7. Verify the connectivity to EC2 instance inside VPC over EC2 private IP

1. Create certificates/keys for Mutual Authentication

- Create Server and Client Certificates and keys
- Upload to ACM



Create Server and Client certificates and keys

Run below commands from your workstation where you have AWS CLI installed (for linux)

1. Clone the easy-rsa repo

```
$ git clone https://github.com/OpenVPN/easy-rsa.git  
$ cd easy-rsa/easyrsa3
```

2. Initialize PKI environment

```
$ ./easyrsa init-pki
```

3. Initialize PKI environment

```
$ ./easyrsa build-ca nopass
```

4. Generate the server certificate and key

```
$ ./easyrsa --san=DNS:server build-server-full server nopass
```

Refer for latest: <https://docs.aws.amazon.com/vpn/latest/clientvpn-admin/client-auth-mutual-enable.html>

Create Server and Client certificates and keys

5. Generate the client certificate and key

```
$ ./easysrsa build-client-full client1.domain.tld nopass
```

6. Copy server and client certificates and keys to one directory

```
$ mkdir ~/demo
$ cp pki/ca.crt ~/demo/
$ cp pki/issued/server.crt ~/demo/
$ cp pki/private/server.key ~/demo/
$ cp pki/issued/client1.domain.tld.crt ~/demo/
$ cp pki/private/client1.domain.tld.key ~/demo/
$ cd ~/demo
```

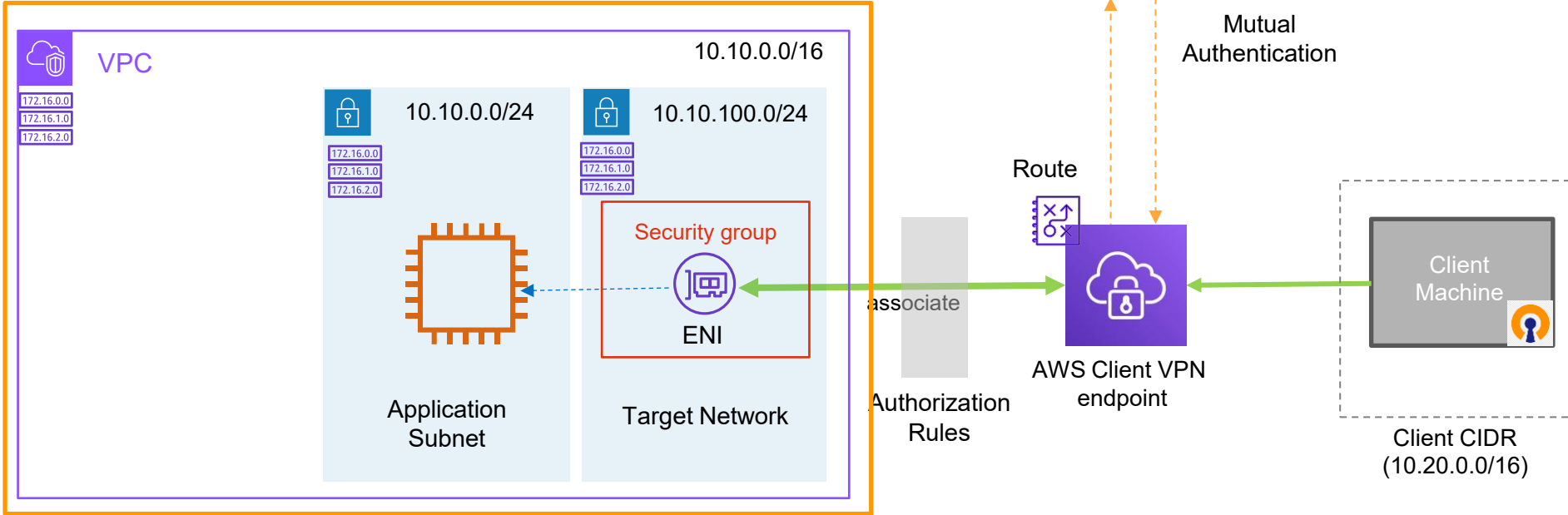
7. Upload the certificate and keys to ACM

```
$ aws acm import-certificate --certificate fileb://server.crt --private-key
fileb://server.key --certificate-chain fileb://ca.crt --region ap-south-1

$ aws acm import-certificate --certificate fileb://client1.domain.tld.crt --private-key
fileb://client1.domain.tld.key --certificate-chain fileb://ca.crt --region ap-south-1
```

2. Setup VPC

- Create VPC and 2 Subnets (private) and route tables
- Create security group for Target Network subnet
- Launch application instance in Application subnet

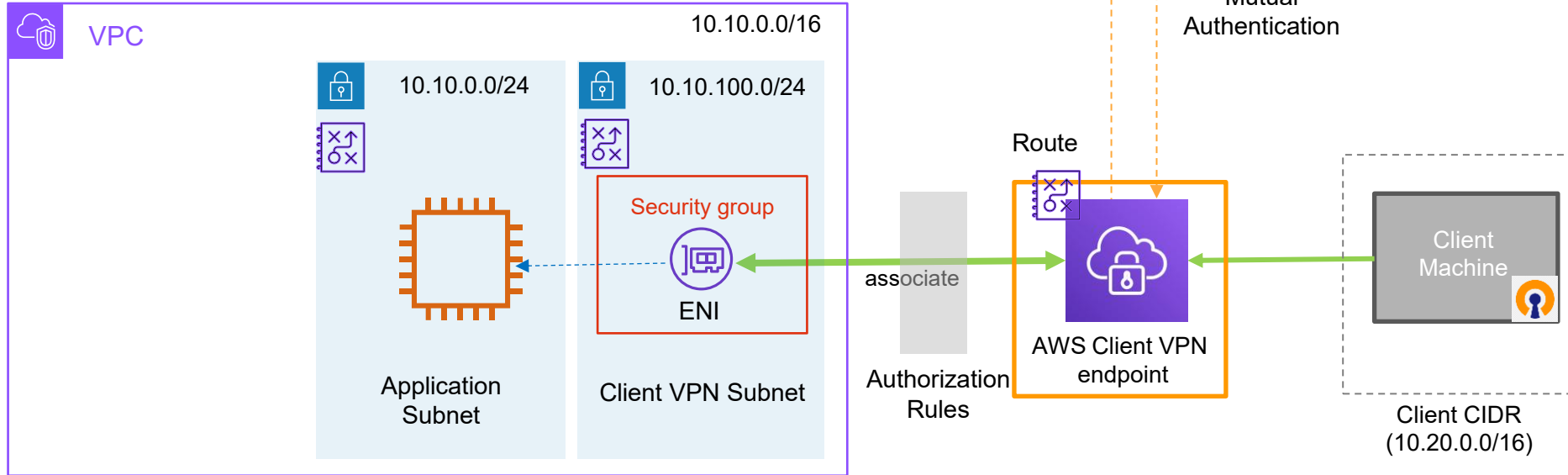


Steps to setup VPC

1. Create VPC with CIDR 10.10.0.0/16
2. Create Private subnet “demo-app” with CIDR 10.10.0.0/24
3. Create corresponding route table “demo-app-rt” with just a local route & associate with subnet “demo-app”
4. Create private subnet “demo-client-vpn” with CIDR 10.10.100.0/24
5. Create corresponding route table “demo-client-vpn-rt” with just a local route & associate with subnet “demo-client-vpn”
6. Create security group “demo-client-vpn-sg”
 - Do not add any inbound rules
 - All outbound should be allowed (All traffic – 0.0.0.0/0)
7. Launch application EC2 instance in “demo-app” subnet
 - Security group inbound rule should allow “All traffic” from security group “demo-client-vpn-sg” created in step 6

3. Create AWS Client VPN Endpoint

- Provide Client CIDR address (10.20.0.0/16)
- Provide ACM Server and Client Certificate
- Provide VPC and Security group details

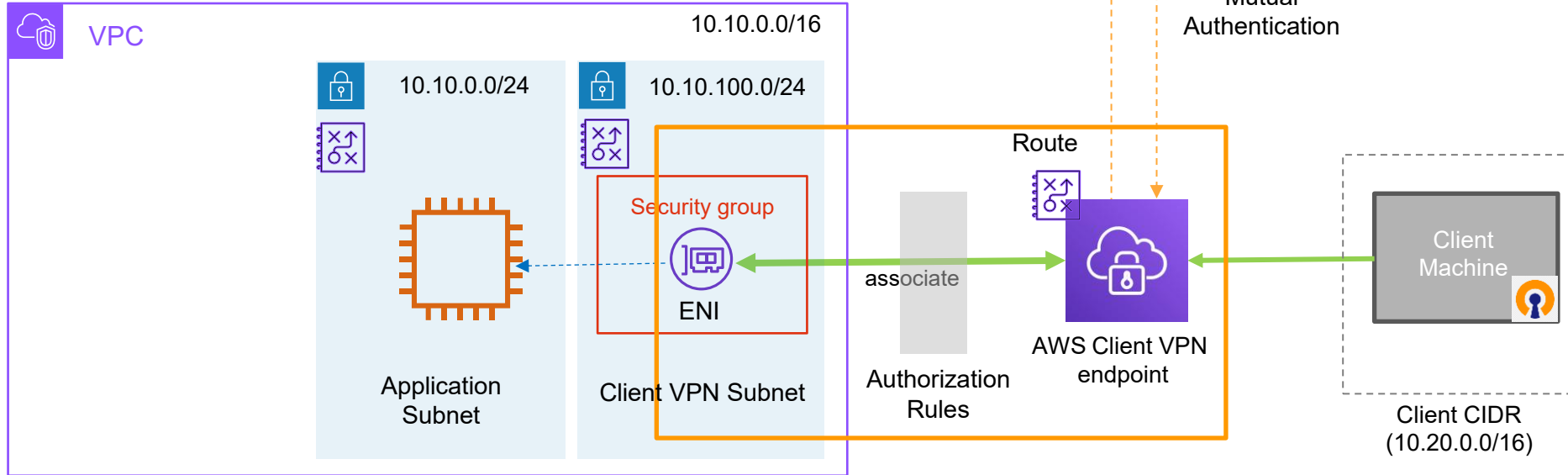


Steps to create Client VPN endpoint

- Provide name “demo-client-vpn-endpoint” and description
- Client IPv4 CIDR: 10.20.0.0/16
- Server Certificate ARN: Choose the Server Certificate created earlier
- Authentication Options: Choose “Use Mutual Authentication”
- Client certificate ARN: Choose the Client Certificate created earlier
- Connection Logging: No
- Transport Protocol: TCP
- VPC ID: Choose “demo” VPC created in Step 2
- Security Group IDs: Select the “demo-client-vpn-sg” created earlier
- VPN port: 443
- Create Client VPN Endpoint

4. Associate Target Subnet & Authorize traffic

- Target subnet (10.10.100.0/24)
- Authorize clients to access the network
- Verify the route for VPC

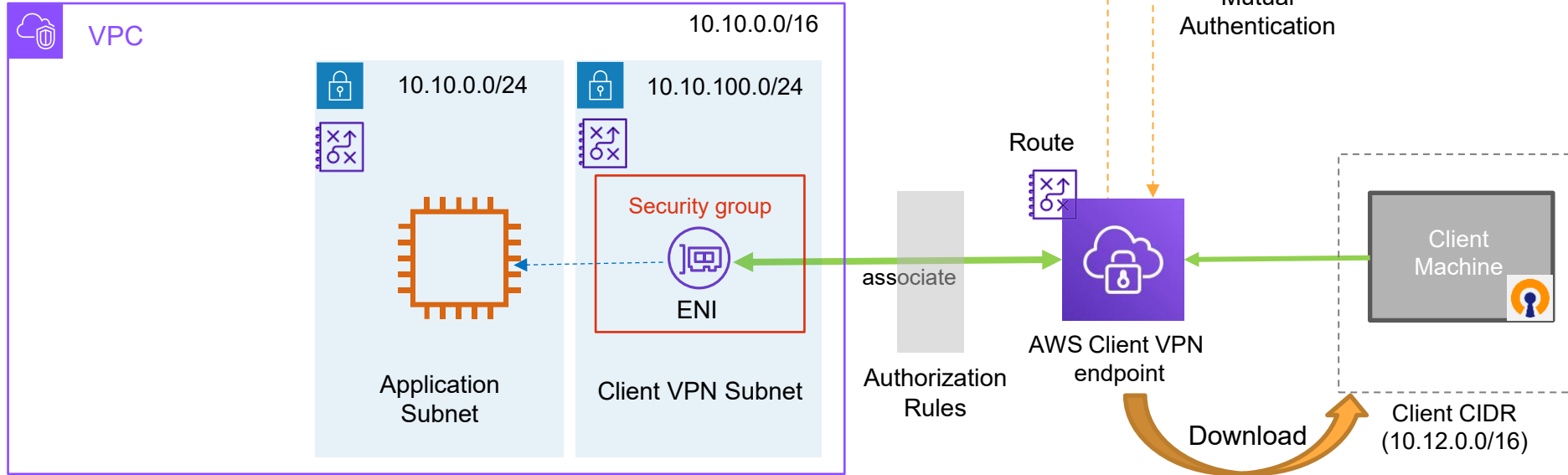


Steps to associate Target Subnet and Authorize traffic

- Select the Client VPN endpoint created earlier
- Go to Associations and associate the target subnet “demo-client-vpn-1”
- Go to Authorizations and choose Authorize Ingress
 - For Destination Networks to enable -> Enter the VPC IP address 10.10.0.0/16
 - Grant access to -> Choose "Allow access to all users"
- Add Authorization Rule

5. Download and update VPN configuration file

- Download client certificate and key to your local machine
- Add the Client Certificate and Key details in the file
- Update the VPN endpoint



Steps to download and update VPN configuration file

- Select Client VPN endpoint and “Download Client Configuration” to your local workstation
- Copy the client certificate and client key created in Step 1 to any folder in local workstation
- Open the configuration file with any editor and add following sections:
- Option 1: Directly add the certificate and key file content under tags

```
<cert>  
Contents of client certificate (.crt) file  
</cert>
```

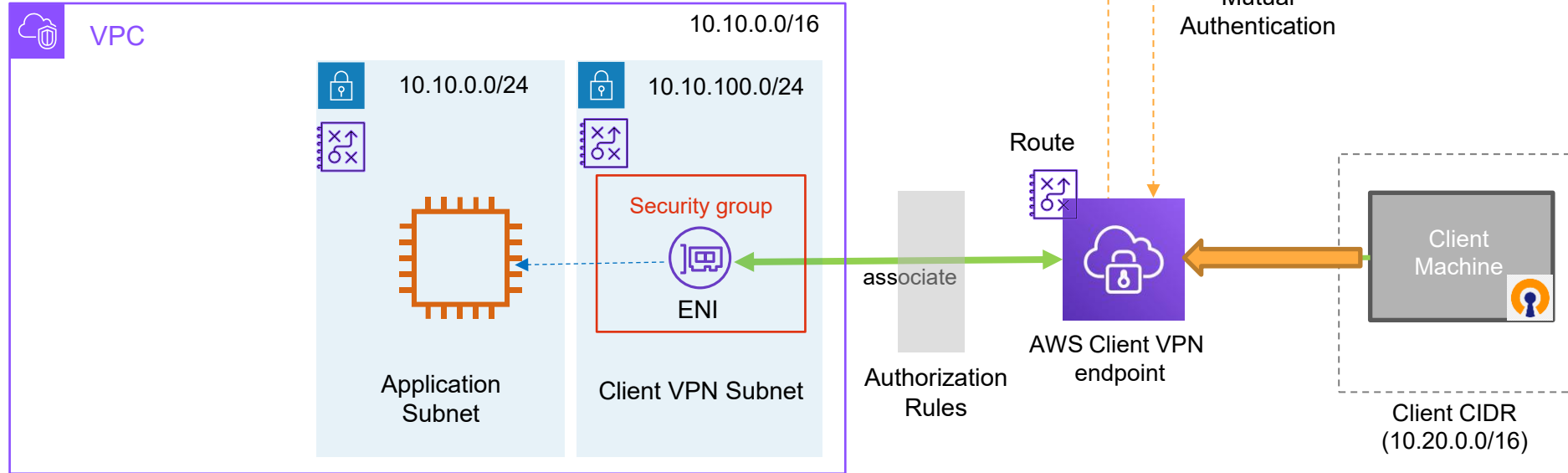
```
<key>  
Contents of private key (.key) file  
</key>
```

- Option 2: Save the cert and key files and add the path as per your local OS

```
cert /path/to/client1.domain.tld.crt  
key /path/to/client1.domain.tld.key
```

6. Connect

- Import the VPN configuration file in OpenVPN client
- Connect

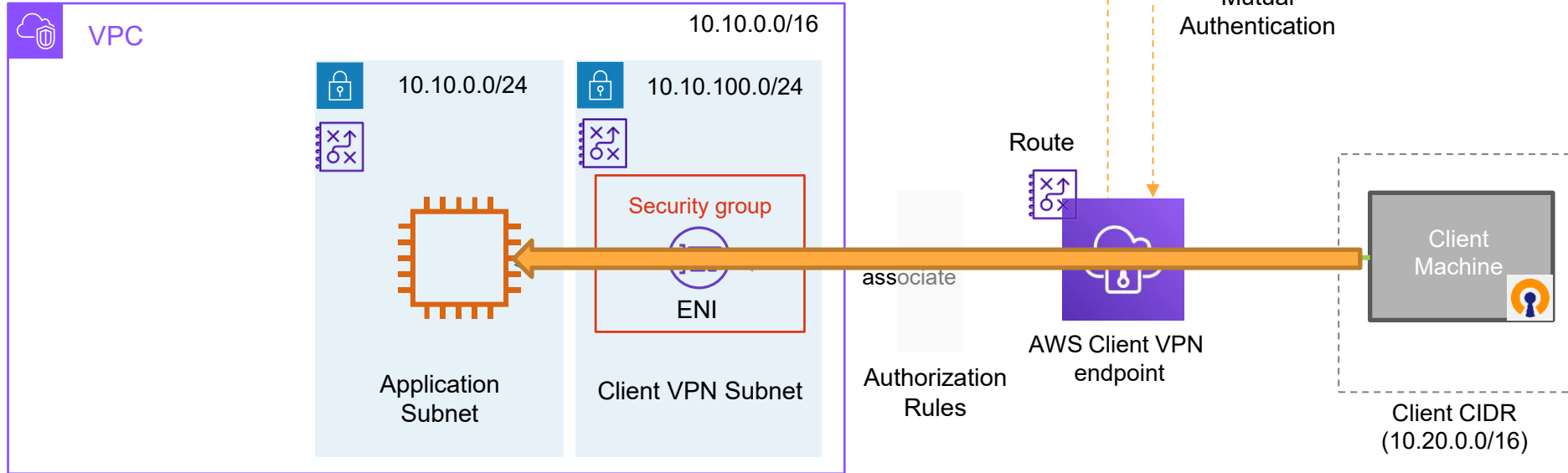


Steps to connect

- Pre-requisite: You should download and install OpenVPN client
 - <https://openvpn.net/community-downloads/>
- Import configuration file
- Connect

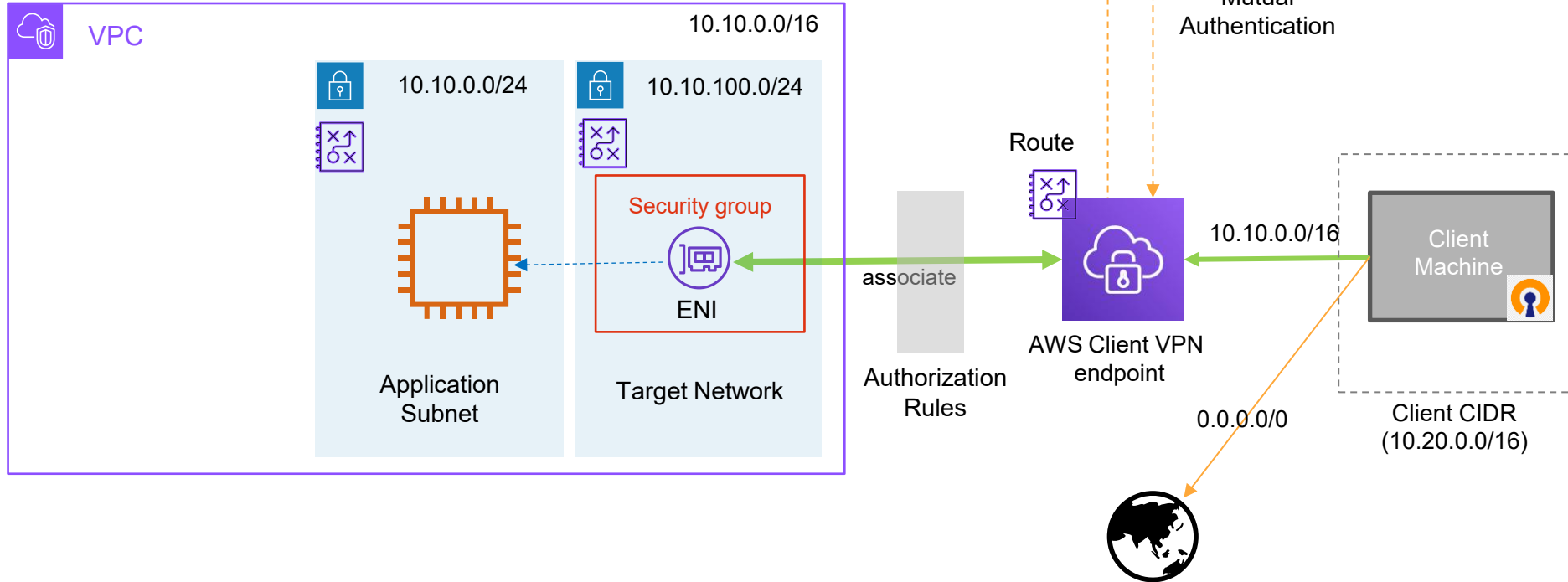
7. Verify the connectivity

- Try to ping or ssh to Application EC2 instance from your local machine



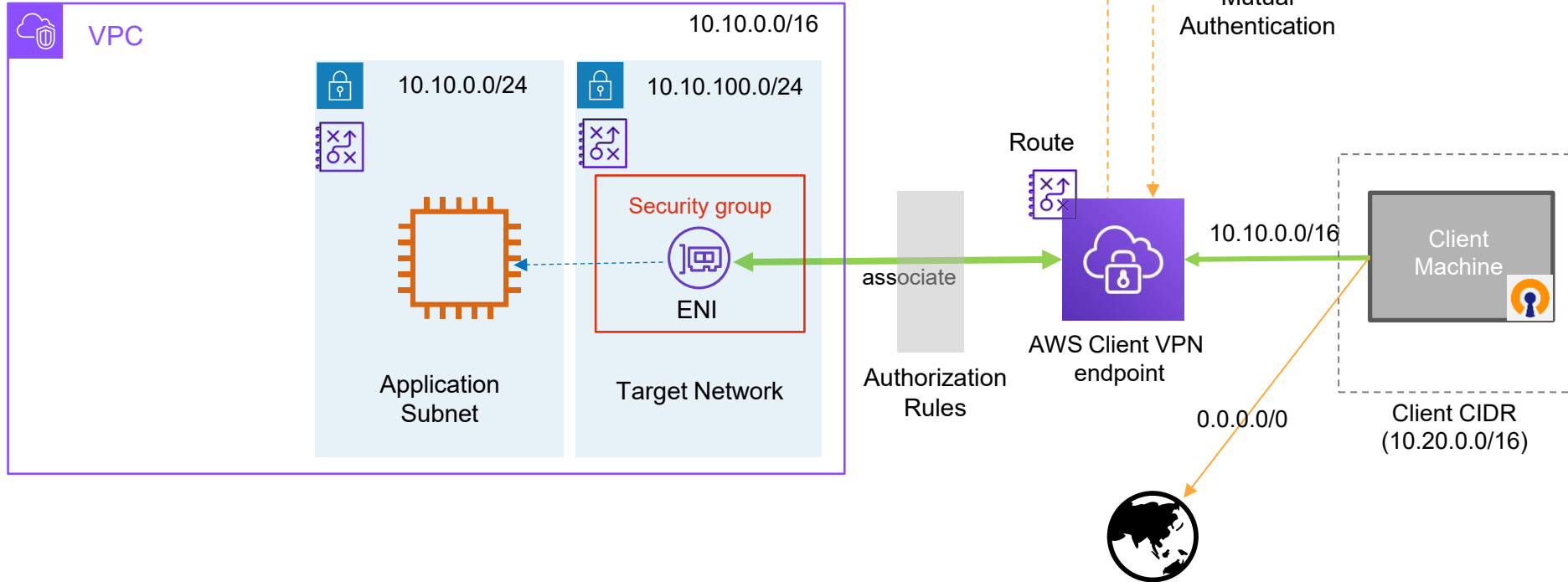
How to Connect to Internet?

- Option 1: Using local internet (Split Tunnel)



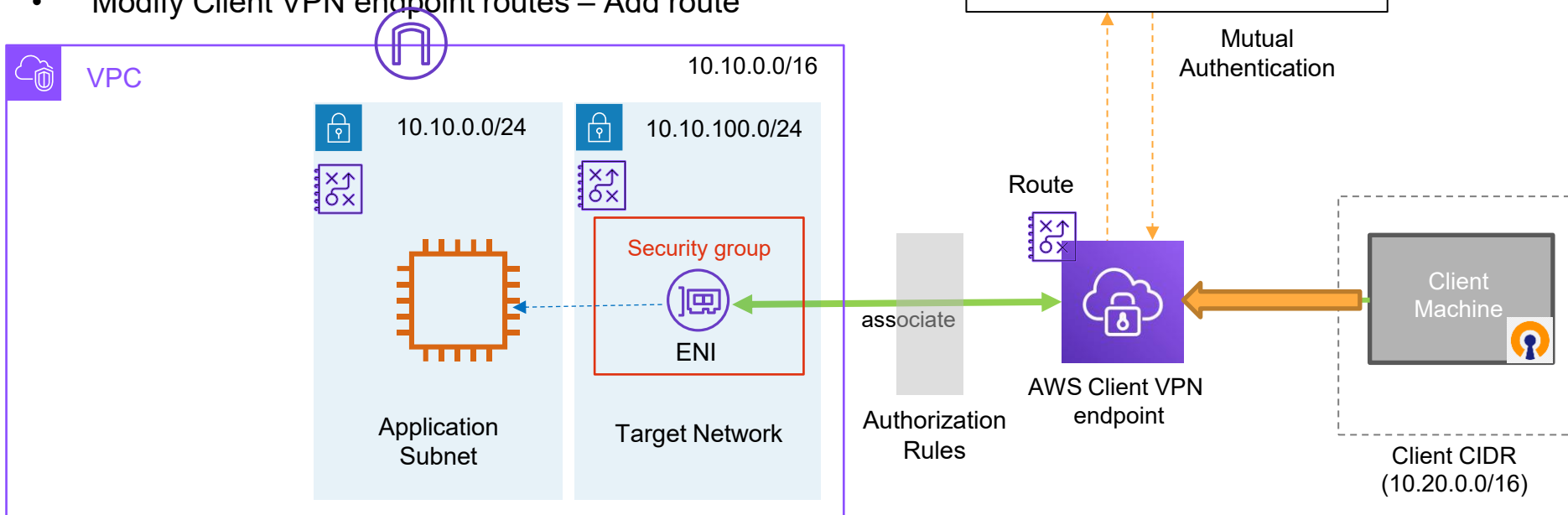
Connecting to Internet using split tunnel

- Disconnect VPN
- Enable Split Tunnel on Client VPN endpoint & Reconnect



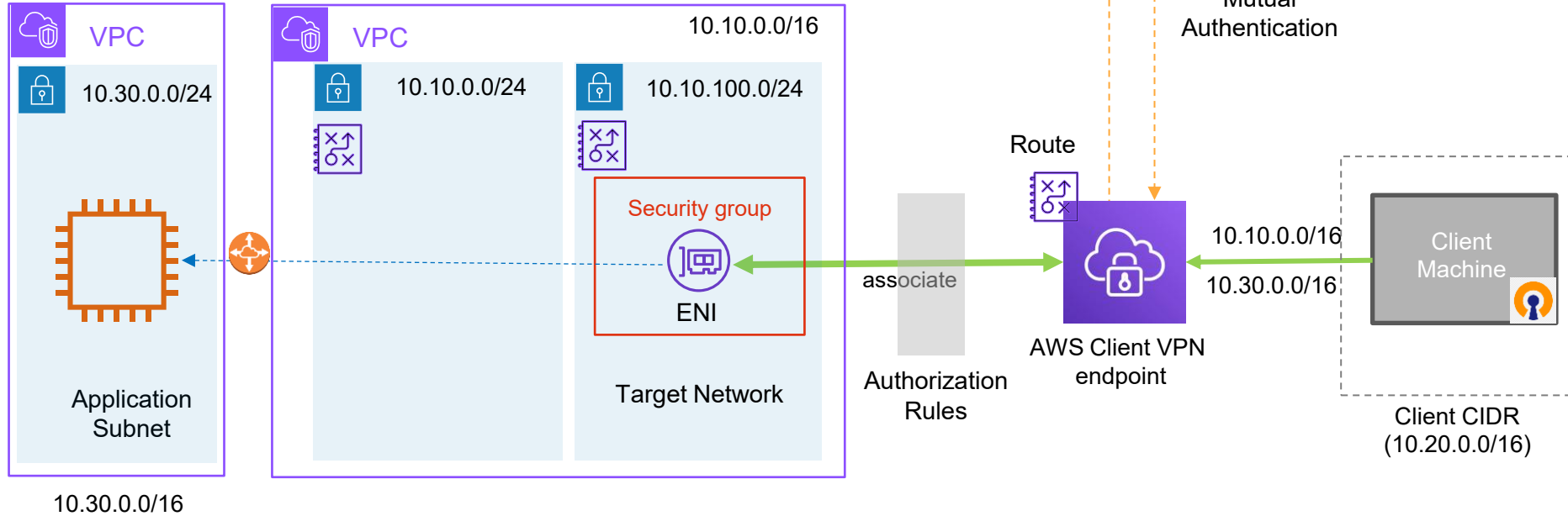
Connecting to Internet using VPC IGW

- Add IGW to your VPC
- Modify Target Subnet Route Table – Add route
- Modify Client VPN endpoint Authorization Rule – Add rule
- Modify Client VPN endpoint routes – Add route



How to Connect to Peered VPC?

- Update Target Subnet route table for peered VPC
- Update Peered VPC resource security group to allow all traffic from target subnet security group
- Add Authorization rule and Client VPN endpoint routes




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